Summer Training Program **WEEK 2 - DAY 4**

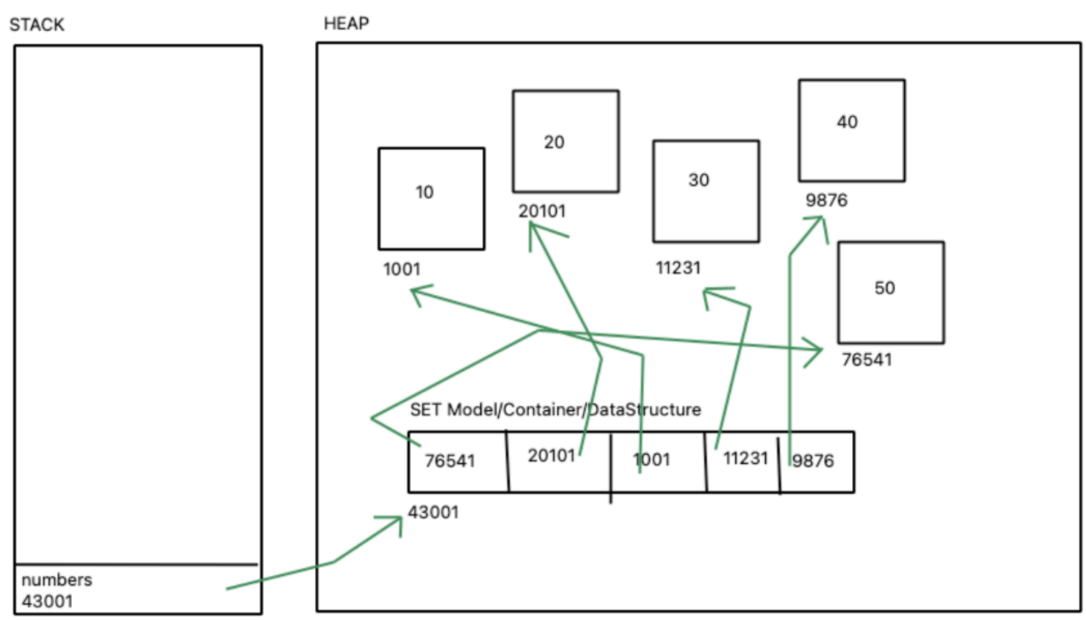
- Set in memory
- Dictionary in memory
- Main frame
- Introduction to Loops

Set: representation in memory

Say there is a set

numbers={10,20,30,40,50}

Representation in memory:



NOTE:

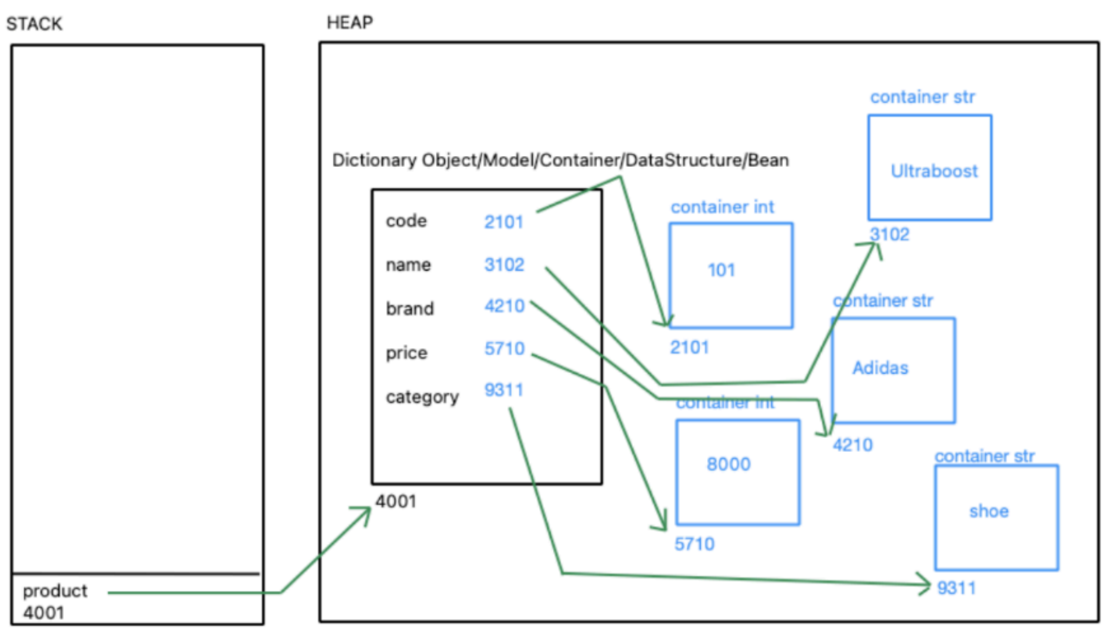
1. How to convert duplicity (list) into uniqueness (set)
    a. Numbers is a list which contains duplicate items
    b. We can use data = set (numbers) —- makes a new container which stores the set and all the duplicate items are not written in it.
2. In c++ , hashcodes can be directly accessed using pointers
3. A container is called a bean during backend development.

Dictionary: Representation in memory

Say there is a dictionary

```
product = {
    'code': 101,
    'name': 'Ultraboost',
    'brand': 'Adidas',
    'price': 8000,
    'category': 'shoe'
}
```

Representation in memory:



Facts:

1. Keys do not store the value directly but store the hashcodes of the values
2. The values are the actual literals/data which cannot be ever changed.
3. We can use hex(), oct() and bin() functions to convert the hashcode into the required number system.

Mainframe: Does a main function exist in python?

- Yes, a main function exists in python, even when we do not define it, it's automatically defined when we write the python code.
- When variables are created they are not pushed into the stack as it is, there is a frame generated in the stack, This is where all the variables of a process are stored.
- A main function will not be executed automatically (like in c++), but it has to be executed by writing a specific code.

Code for creation for main function:

```
def main():
    print ("Hello, This is the mainframe!")
```

Name of the function can be customised

Any statements inside the main have to be written after one tab or 4 tab spaces.

Code for executing main function:

```
if __name__ == "__main__":
    main()
```

The main function can be executed using this code. This is the industrial standard and more formal way of execution.

OR

```
main()
```

This is also an informal way of executing main function.

Output:

```
Hello, This is the mainframe!
```

Facts:

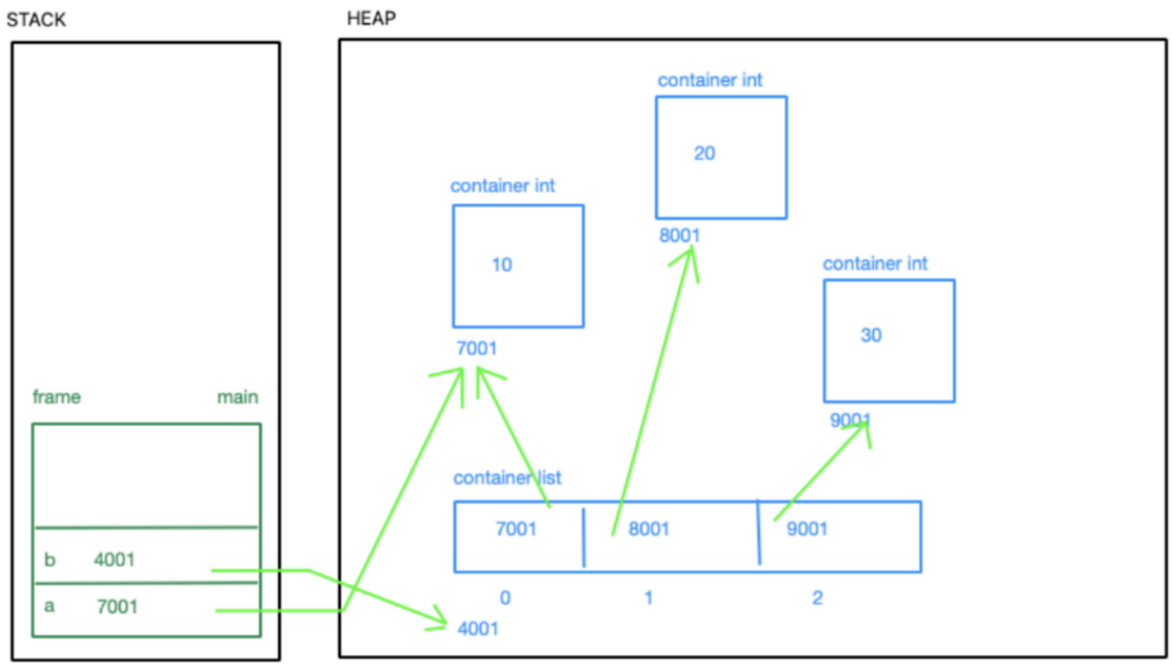
1. If some code is present outside main, it is called global code.
2. Data is not dumped into the stack directly, else it is dumped in a frame.
3. A frame is a memory block allotted to the process.
4. A frame belongs to the main function.
5. It does not matter if we make a main function or not, a frame will always be generated in the stack.
6. Python source code is called a module.
7. Main thread is conceptually equivalent to the main frame.

Mainframe in memory

CODE:

```
a = 10
b = {10, 20, 30}
```

Memory representation:



This is the correct memory representation of variables inside a stack.
Hence, all the diagrams discussed forth with reflect the same change in stack representation.

Loops:

Loops help in iterating and performing operations on variables.

1. FOR EACH LOOP/ENHANCED FOR LOOP:

   Say we have a set

   ```
   data=[10,20,30,40,50]
   for number in data:
       print(number)
   ```

   This cannot be used for manipulation. Hence, it can not be used for writing data. It is a read only loop.

2. ENHANCED FOR LOOP:

   ```
   for index in range(0,5):
       print(index)
   ```

   This loop does not work on sets, because sets are not indexed. This loop can also be used for data manipulation. It is a read and write loop.

   range (start, size, increment)
   Start: start index of loop
   Size: size-1 will be up until which iteration the loop will end

3. FOR LOOP (for dictionary) :

   ```
   numbers = {1: "one", 2: "two", 3: "three"}
   for key in numbers:
       print (key, numbers[key])
   ```

   This will print the keys and the values of the dictionary.

4. WHILE LOOP:

   ```
   i=0
   while i<5:
       print(i)
       i+=1
   ```

   Prints numbers 0 to 4
   An increment operator(++) does not exist in python. So we have to manually add one to the value of i to increment it.

Important facts:

1. When we have dimensions of data, we use loops in loops
2. As to what is the process of choosing these loops/logic, we have to optimize two parameters:
    a. Time: Solution should take the least amount of time to reach the desired solution.
    b. Space: Solution should take the least amount of memory space to reach the desired solution. ‘

3. We can customize the end of the print statement,

By default, end = ‘\n’ #newline character
We can also end it with, end = ‘ ’ #space
Also, end= ‘*’ #any other character

INTERVIEW QUESTIONS:

- Which is more important, class or object?
- Is there a main function in python

ASSIGNMENT: Write the code to generate a chessboard in python, use all the specific unicodes for all the chess keys.
(pushed to github)

GURASSIS KAUR
g.assiskaur@gmail.com